

2024 年全国硕士研究生招生考试数学三解析

一、选择题

1. 答 应选 D.

解 当 $|x| < 1$ 时, $\lim_{n \rightarrow \infty} \frac{1+x}{1+nx^{2n}} = 1+x$;

当 $|x| > 1$ 时, $\lim_{n \rightarrow \infty} \frac{1+x}{1+nx^{2n}} = 0$;

当 $x = 1$ 时, $\lim_{n \rightarrow \infty} \frac{2}{1+n} = 0$;

当 $x = -1$ 时, $\lim_{n \rightarrow \infty} \frac{0}{1+n} = 0$.

故 $f(x) = \begin{cases} 1+x, & -1 < x < 1, \\ 0, & x \leq -1 \text{ 或 } x \geq 1. \end{cases}$ 则 $f(x)$ 在 $x = -1$ 处连续, 在 $x = 1$ 处不连续. 故选 D.

2. 答 应选 B.

解 由于 $|\sin x|$ 以 π 为周期, 则 $I = \int_a^{a+k\pi} |\sin x| dx = k \int_0^\pi \sin x dx = 2k$. 故选 B.

3. 答 应选 A.

解 积分区域 D 如图所示, $\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} dx \int_{\sin x}^1 f(x, y) dy = \int_{\frac{1}{2}}^1 dy \int_{\frac{\pi}{6}}^{\arcsin y} f(x, y) dx$.

故选 A.

4. 答 应选 A.

解 $\ln(2+x) = \ln\left(1+\frac{x}{2}\right) + \ln 2 = \ln 2 + \sum_{n=1}^{\infty} (-1)^{n-1} \frac{\left(\frac{x}{2}\right)^n}{n}$
 $= \ln 2 + \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n \cdot 2^n} x^n, -2 < x \leq 2$.

依题意, 有 $a_0 = \ln 2, a_n = \frac{(-1)^{n-1}}{n \cdot 2^n} (n = 1, 2, \dots)$, 进而 $a_{2n} = \frac{(-1)^{2n-1}}{2n \cdot 2^{2n}} = -\frac{1}{2n \cdot 4^n}, n = 1, 2, \dots$.

$$\sum_{n=0}^{\infty} n a_{2n} = \sum_{n=1}^{\infty} n a_{2n} = -\sum_{n=1}^{\infty} n \cdot \frac{1}{2n \cdot 4^n} = -\frac{1}{2} \sum_{n=1}^{\infty} \frac{1}{4^n} = -\frac{1}{2} \cdot \frac{\frac{1}{4}}{1 - \frac{1}{4}} = -\frac{1}{6}.$$

故选 A.

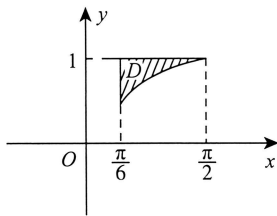
5. 答 应选 C.

解 因为 $f(x_1, x_2, x_3) = \mathbf{x}^T \mathbf{A} \mathbf{x}$ 在正交变换下可化为 $y_1^2 - 2y_2^2 + 3y_3^2$, 所以 $\mathbf{A}$ 的特征值为 $1, -2, 3$, 得
 $|\mathbf{A}| = 1 \times (-2) \times 3 = -6, \operatorname{tr}(\mathbf{A}) = 1 + (-2) + 3 = 2$.

6. 答 应选 C.

解 记 $\mathbf{P}^T \mathbf{A} \mathbf{P}^2 = \begin{pmatrix} a+2c & 0 & c \\ 0 & b & 0 \\ 2c & 0 & c \end{pmatrix} = \mathbf{B}$, 且 $\mathbf{P} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} = \mathbf{E}_{31}(1)$, 故

$$\mathbf{A} = (\mathbf{P}^T)^{-1} \mathbf{B} (\mathbf{P}^2)^{-1} = [\mathbf{E}_{31}^T(1)]^{-1} \mathbf{B} [\mathbf{E}_{31}^2(1)]^{-1}$$



$$\begin{aligned}
&= [\mathbf{E}_{31}^{-1}(1)]^T \mathbf{B} \mathbf{E}_{31}^{-1}(1) \mathbf{E}_{31}^{-1}(1) = \mathbf{E}_{31}^T(-1) \mathbf{B} \mathbf{E}_{31}(-1) \mathbf{E}_{31}(-1) \\
&= \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} a+2c & 0 & c \\ 0 & b & 0 \\ 2c & 0 & c \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix} \\
&= \begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 2c & 0 & c \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix} = \begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{pmatrix}.
\end{aligned}$$

7. 答 应选 B.

解 由题意知, $|\mathbf{A}| = \begin{vmatrix} a+1 & b & 3 \\ a & \frac{b}{2} & 1 \\ 1 & 1 & 2 \end{vmatrix} = \begin{vmatrix} 1 & \frac{b}{2} & 2 \\ a & \frac{b}{2} & 1 \\ 1 & 1 & 2 \end{vmatrix} = \begin{vmatrix} 0 & \frac{b}{2}-1 & 0 \\ a & \frac{b}{2} & 1 \\ 1 & 1 & 2 \end{vmatrix} = -\frac{1}{2}$, 则

$$\left(\frac{b}{2}-1\right)(2a-1) = \frac{1}{2}.$$

整理得

$$ab - 2a - \frac{b}{2} + 1 = \frac{1}{2}. \quad (*)$$

又

$$-M_{21} + M_{22} - M_{23} = A_{21} + A_{22} + A_{23} = \begin{vmatrix} a+1 & b & 3 \\ 1 & 1 & 1 \\ 1 & 1 & 2 \end{vmatrix} = \begin{vmatrix} a+1 & b & 3 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{vmatrix} = a+1-b=0,$$

则有 $b = a+1$, 代入(*)式中, 得 $a(a+1) - 2a - \frac{a+1}{2} + \frac{1}{2} = 0$, 解得 $a = 0$ 或 $a = \frac{3}{2}$, 从而 $b = 1$ 或 $\frac{5}{2}$.

8. 答 应选 B.

解 $EX = \int_0^1 6x^2(1-x)dx = 6 \times \left(\frac{1}{3} - \frac{1}{4}\right) = 6 \times \frac{1}{12} = \frac{1}{2}$, 则

$$E(X-EX)^3 = E\left(X - \frac{1}{2}\right)^3 = \int_0^1 6x(1-x) \left(x - \frac{1}{2}\right)^3 dx$$

$$\stackrel{\text{令 } x - \frac{1}{2} = t}{=} \int_{-\frac{1}{2}}^{\frac{1}{2}} 6\left(t + \frac{1}{2}\right) \cdot \left(\frac{1}{2} - t\right) \cdot t^3 dt = 0.$$

9. 答 应选 B.

解

$$E(2X-Y) = 2EX - EY = 0+1 = 1,$$

$$D(2X-Y) = 4DX + DY = 4 \times 2 + 1 = 9,$$

所以 $2X-Y \sim N(1, 9)$;

$$E(X-2Y) = EX - 2EY = 0+2 = 2,$$

$$D(X-2Y) = DX + 4DY = 2+4 = 6,$$

所以 $X-2Y \sim N(2, 6)$.

故 $p_1 = P\{2X > Y\} = P\left\{\frac{2X-Y-1}{3} > \frac{0-1}{3}\right\} = 1 - \Phi\left(-\frac{1}{3}\right) = \Phi\left(\frac{1}{3}\right),$

$$p_2 = P\{X-2Y > 1\} = P\left\{\frac{X-2Y-2}{\sqrt{6}} > \frac{1-2}{\sqrt{6}}\right\} = 1 - \Phi\left(-\frac{1}{\sqrt{6}}\right) = \Phi\left(\frac{1}{\sqrt{6}}\right),$$

所以 $p_2 > p_1 > \frac{1}{2}$, 故选 B.

10. 答 应选 D.

解 X 与 Y 的联合概率密度为 $f(x, y) = f_X(x) \cdot f_Y(y) = \begin{cases} \lambda^2 e^{-\lambda(x+y)}, & x > 0, y > 0, \\ 0, & \text{其他.} \end{cases}$

设 Z 的分布函数为 $F_Z(z)$, 则 $F_Z(z) = P\{Z \leq z\} = P\{|X - Y| \leq z\}$.

当 $z < 0$ 时, $F_Z(z) = 0$;

当 $z \geq 0$ 时,

$$\begin{aligned} F_Z(z) &= P\{-z \leq X - Y \leq z\} = 2P\{0 \leq X - Y \leq z\} \\ &= 2 \int_0^{+\infty} \lambda e^{-\lambda y} dy \int_y^{y+z} \lambda e^{-\lambda x} dx \\ &= 2 \int_0^{+\infty} \lambda e^{-\lambda y} [e^{-\lambda y} - e^{-\lambda(y+z)}] dy \\ &= 2 \int_0^{+\infty} \lambda e^{-2\lambda y} dy - 2e^{-\lambda z} \int_0^{+\infty} \lambda e^{-2\lambda y} dy \\ &= 1 - e^{-\lambda z}. \end{aligned}$$

所以 $Z \sim E(\lambda)$, 从而 Z 与 X 服从相同的分布, 选 D.

二、填空题

11. 答 应填 3.

解 当 $x \rightarrow 0$ 时, $\frac{(1+x^2)\sin x^2}{1+\cos^2 x} \sim \frac{x^2}{2}$, 则 $\int_0^x \frac{(1+t^2)\sin t^2}{1+\cos^2 t} dt \sim \int_0^x \frac{t^2}{2} dt = \frac{1}{6}x^3$. 从而 $k = 3$.

12. 答 应填 $\frac{1}{2} \ln 3 - \frac{\pi}{8}$.

$$\begin{aligned} \text{解} \quad \int_2^{+\infty} \frac{5}{x^4 + 3x^2 - 4} dx &= \int_2^{+\infty} \frac{5}{(x^2 - 1)(x^2 + 4)} dx = \int_2^{+\infty} \frac{1}{x^2 - 1} dx - \int_2^{+\infty} \frac{1}{x^2 + 4} dx \\ &= \frac{1}{2} \ln \frac{x-1}{x+1} \Big|_2^{+\infty} - \frac{1}{2} \arctan \frac{x}{2} \Big|_2^{+\infty} = \frac{1}{2} (0 - \ln \frac{1}{3}) - \frac{1}{2} (\frac{\pi}{2} - \frac{\pi}{4}) \\ &= \frac{1}{2} \ln 3 - \frac{\pi}{8}. \end{aligned}$$

13. 答 应填 (1, 1).

解 令 $\begin{cases} f'_x = 6x^2 - 18x + 12 = 0, \\ f'_y = -24y^3 + 24 = 0, \end{cases}$ 解得驻点为 (1, 1), (2, 1).

$$A = f''_{xx} = 12x - 18, B = f''_{xy} = 0, C = f''_{yy} = -72y^2.$$

代入点 (1, 1) 得 $AC - B^2 = 432 > 0, A = -6 < 0$, 故 (1, 1) 是极大值点;

代入点 (2, 1) 得 $AC - B^2 = -432 < 0$, 故 (2, 1) 不是极值点.

14. 答 应填 50.

$$\text{解} \quad L = pQ - C = \begin{cases} (25 - 0.25Q)Q - (150 + 5Q + 0.25Q^2), & Q \leq 20, \\ (35 - 0.75Q)Q - (150 + 5Q + 0.25Q^2), & Q > 20. \end{cases}$$

整理得 $L = \begin{cases} -0.5(Q - 20)^2 + 50, & Q \leq 20, \\ -(Q - 15)^2 + 75, & Q > 20. \end{cases}$ 所以 $Q = 20$ 时, $L = 50$ 为最大利润.

15. 答 应填 16.

解 $r(2E - A) = 1$, 则 $(2E - A)x = 0$ 有两个线性无关解, 于是 $\lambda = 2$ 有 2 个线性无关的特征向量, 即 $\lambda = 2$ 至少为二重特征值.

$r(E + A) = r(-E - A) = 2$, 则 $(-E - A)x = 0$ 有一个线性无关解, 于是 $\lambda = -1$ 有 1 个线性无关特征向量, 即 $\lambda = -1$ 至少为一重特征值.

又 A 为 3 阶矩阵, 所以 A 的特征值为 2, 2, -1. 故 $|A| = 2 \times 2 \times (-1) = -4$, $|A^*| = |A|^{n-1} = |A|^2 = 16$.

16. 答 应填 $\frac{2}{3}$.

解 设事件 A 为 3 次试验全部成功, 事件 B 为至少成功 1 次, 则 $P(A|B) = \frac{P(AB)}{P(B)} = \frac{P(A)}{P(B)} = \frac{p^3}{1-(1-p)^3} = \frac{4}{13}$, 于是 $13p^3 = 4 - 4(1-p)^3$, 整理得 $p(3p-2)(3p+6) = 0$, 解得 $p = \frac{2}{3}$.

三、解答题

17. 解 由于 D 关于直线 $y = x$ 对称, 因此 $\iint_D x dx dy = \iint_D y dx dy$.

令 $D_1 = \left\{ (x, y) \mid \frac{1}{3x} \leq y \leq 3x, \frac{1}{3} \leq x \leq 1 \right\}$, $D_2 = \left\{ (x, y) \mid \frac{x}{3} \leq y \leq \frac{3}{x}, 1 \leq x \leq 3 \right\}$.

$$\begin{aligned} \iint_D (1+x-y) dx dy &= \iint_D dx dy = \iint_{D_1} dx dy + \iint_{D_2} dx dy \\ &= \int_{\frac{1}{3}}^1 dx \int_{\frac{1}{3x}}^{3x} dy + \int_1^3 dx \int_{\frac{x}{3}}^{\frac{3}{x}} dy \\ &= \int_{\frac{1}{3}}^1 \left(3x - \frac{1}{3x} \right) dx + \int_1^3 \left(\frac{3}{x} - \frac{x}{3} \right) dx \\ &= \left(\frac{3}{2} x^2 - \frac{1}{3} \ln x \right) \Big|_{\frac{1}{3}}^1 + \left(3 \ln x - \frac{1}{6} x^2 \right) \Big|_1^3 \\ &= \frac{8}{3} \ln 3. \end{aligned}$$

18. 解 将 $x = 0, y = 0$ 代入方程 $z + e^x - y \ln(1+z^2) = 0$, 得 $z(0,0) = -1$.

在方程 $z + e^x = y \ln(1+z^2)$ 的两边对 x 求偏导数, 得 $\frac{\partial z}{\partial x} + e^x = \frac{2yz}{1+z^2} \frac{\partial z}{\partial x}$, 因此 $\frac{\partial z}{\partial x} \Big|_{(0,0)} = -1$.

在方程 $z + e^x = y \ln(1+z^2)$ 的两边对 y 求偏导数, 得 $\frac{\partial z}{\partial y} = \ln(1+z^2) + \frac{2yz}{1+z^2} \frac{\partial z}{\partial y}$, 因此 $\frac{\partial z}{\partial y} \Big|_{(0,0)} = \ln 2$.

在方程 $\frac{\partial z}{\partial x} + e^x = \frac{2yz}{1+z^2} \frac{\partial z}{\partial x}$ 的两边对 x 求偏导数, 得 $\frac{\partial^2 z}{\partial x^2} + e^x = \frac{2yz}{1+z^2} \frac{\partial^2 z}{\partial x^2} + \frac{2y(1-z^2)}{(1+z^2)^2} \left(\frac{\partial z}{\partial x} \right)^2$, 因此

$$\frac{\partial^2 z}{\partial x^2} \Big|_{(0,0)} = -1.$$

在方程 $\frac{\partial z}{\partial y} = \ln(1+z^2) + \frac{2yz}{1+z^2} \frac{\partial z}{\partial y}$ 的两边对 y 求偏导数, 得 $\frac{\partial^2 z}{\partial y^2} = \frac{4z}{1+z^2} \frac{\partial z}{\partial y} + \frac{2y(1-z^2)}{(1+z^2)^2} \left(\frac{\partial z}{\partial y} \right)^2 +$

$$\frac{2yz}{1+z^2} \frac{\partial^2 z}{\partial y^2}, \text{ 因此 } \frac{\partial^2 z}{\partial y^2} \Big|_{(0,0)} = -2 \ln 2.$$

综上所述, $\left(\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} \right) \Big|_{(0,0)} = -1 - 2 \ln 2$.

19. 解 D 的面积 $S(t) = \int_t^{2t} x e^{-2x} dx (t > 0)$, 则 $S'(t) = 4t e^{-4t} - t e^{-2t} = -t e^{-4t} (e^{2t} - 4)$. 令 $S'(t) = 0$,

得 $t = \ln 2$.

当 $0 < t < \ln 2$ 时, $S'(t) > 0$; 当 $t > \ln 2$ 时, $S'(t) < 0$. 所以 $S(t)$ 在 $(0, \ln 2)$ 内单调递增, 在 $(\ln 2, +\infty)$ 内单调递减, 故当 $t = \ln 2$ 时 $S(t)$ 最大.

$S(t)$ 的最大值为

$$\begin{aligned} S(\ln 2) &= \int_{\ln 2}^{2 \ln 2} x e^{-2x} dx = -\frac{1}{2} x e^{-2x} \Big|_{\ln 2}^{2 \ln 2} + \frac{1}{2} \int_{\ln 2}^{2 \ln 2} e^{-2x} dx \\ &= \frac{\ln 2}{16} - \frac{1}{4} e^{-2x} \Big|_{\ln 2}^{2 \ln 2} \\ &= \frac{\ln 2}{16} + \frac{3}{64}. \end{aligned}$$

20. 证 (1) 因为 $f(x)$ 具有 2 阶导数, 当 $x \in (0, 1)$ 时, 根据泰勒公式, 存在 $\xi_1, \xi_2 \in (0, 1)$, 使得

$$f(x) = f(0) + f'(0)x + \frac{1}{2} f''(\xi_1) x^2,$$

$$f(x) = f(1) + f'(1)(x-1) + \frac{1}{2}f''(\xi_2)(x-1)^2.$$

所以 $(x-1)f(x) = f(0)(x-1) + f'(0)x(x-1) + \frac{1}{2}f''(\xi_1)x^2(x-1),$

$$xf(x) = f(1)x + f'(1)(x-1)x + \frac{1}{2}f''(\xi_2)(x-1)^2x.$$

因为 $f'(0) = f'(1)$, 所以

$$f(x) = f(0)(1-x) + f(1)x + \frac{x(1-x)}{2}[xf''(\xi_1) + (1-x)f''(\xi_2)].$$

又因为 $|f''(x)| \leq 1$, 所以当 $x \in (0, 1)$ 时,

$$\begin{aligned} & |f(x) - f(0)(1-x) - f(1)x| \\ & \leq \frac{x(1-x)}{2}[x|f''(\xi_1)| + (1-x)|f''(\xi_2)|] \leq \frac{x(1-x)}{2}. \end{aligned}$$

(2) 因为 $\int_0^1 [f(x) - f(0)(1-x) - f(1)x] dx = \int_0^1 f(x) dx - \frac{f(0) + f(1)}{2}$, 所以由定积分的性质和(1)

可得

$$\begin{aligned} \left| \int_0^1 f(x) dx - \frac{f(0) + f(1)}{2} \right| & \leq \int_0^1 |f(x) - f(0)(1-x) - f(1)x| dx \\ & \leq \int_0^1 \frac{x(1-x)}{2} dx = \frac{1}{12}. \end{aligned}$$

21. (1) 证 对 $Ax = \alpha$ 的增广矩阵施以初等行变换, 有

$$(A : \alpha) = \left(\begin{array}{cccc|c} 1 & -1 & 0 & -1 & 0 \\ 1 & 1 & 0 & 3 & 2 \\ 2 & 1 & 2 & 6 & 3 \end{array} \right) \rightarrow \left(\begin{array}{cccc|c} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 2 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{array} \right),$$

得 $Ax = \alpha$ 的通解为

$$x = \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix} + k \begin{pmatrix} -1 \\ -2 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1-k \\ 1-2k \\ -k \\ k \end{pmatrix}, \text{ 其中 } k \text{ 为任意常数.}$$

将 $Ax = \alpha$ 的通解代入 Bx 得

$$Bx = \begin{pmatrix} 1 & 0 & 1 & 2 \\ 1 & -1 & a & a-1 \\ 2 & -3 & 2 & -2 \end{pmatrix} \begin{pmatrix} 1-k \\ 1-2k \\ -k \\ k \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} = \beta,$$

所以方程组 $Ax = \alpha$ 的解均为方程组 $Bx = \beta$ 的解.

(2) 解 对 $Bx = \beta$ 的增广矩阵施以初等行变换, 有

$$(B : \beta) = \left(\begin{array}{cccc|c} 1 & 0 & 1 & 2 & 1 \\ 1 & -1 & a & a-1 & 0 \\ 2 & -3 & 2 & -2 & -1 \end{array} \right) \rightarrow \left(\begin{array}{cccc|c} 1 & 0 & 1 & 2 & 1 \\ 0 & 1 & 1-a & 3-a & 1 \\ 0 & 0 & a-1 & a-1 & 0 \end{array} \right).$$

当 $a \neq 1$ 时, 再经初等行变换, 有

$$(B : \beta) \rightarrow \left(\begin{array}{cccc|c} 1 & 0 & 1 & 2 & 1 \\ 0 & 1 & 1-a & 3-a & 1 \\ 0 & 0 & a-1 & a-1 & 0 \end{array} \right) \rightarrow \left(\begin{array}{cccc|c} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 2 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{array} \right),$$

得方程组 $Bx = \beta$ 的通解为

$$\mathbf{x} = \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix} + k \begin{pmatrix} -1 \\ -2 \\ -1 \\ 1 \end{pmatrix}, \text{其中 } k \text{ 为任意常数,}$$

所以方程组 $\mathbf{Ax} = \boldsymbol{\alpha}$ 与方程组 $\mathbf{Bx} = \boldsymbol{\beta}$ 同解.

当 $a = 1$ 时,经初等行变换,有

$$(\mathbf{B} : \boldsymbol{\beta}) \rightarrow \left(\begin{array}{cccc|c} 1 & 0 & 1 & 2 & 1 \\ 0 & 1 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right),$$

知 $r(\mathbf{A} : \boldsymbol{\alpha}) \neq r(\mathbf{B} : \boldsymbol{\beta})$, 所以方程组 $\mathbf{Ax} = \boldsymbol{\alpha}$ 与方程组 $\mathbf{Bx} = \boldsymbol{\beta}$ 不同解.

综上可知, $a = 1$.

$$22. \text{解} \quad (1) X \text{ 的分布函数为 } F(x) = \begin{cases} 0, & x < 0, \\ \frac{x}{\theta}, & 0 \leq x < \theta, \\ 1, & x \geq \theta, \end{cases} \text{则 } X_{(n)} \text{ 的概率密度为}$$

$$f_{(n)}(x) = n[F(x)]^{n-1}F'(x) = \begin{cases} \frac{n}{\theta^n}x^{n-1}, & 0 \leq x \leq \theta, \\ 0, & \text{其他,} \end{cases}$$

$$\text{从而 } EX_{(n)} = \int_{-\infty}^{+\infty} xf_{(n)}(x)dx = \int_0^\theta \frac{n}{\theta^n}x^n dx = \frac{n}{n+1}\theta.$$

$$\text{故 } ET_c = \frac{cn}{n+1}\theta, \text{所以当 } c = \frac{n+1}{n} \text{ 时, } ET_c = \theta.$$

(2) 由(1) 知

$$EX_{(n)}^2 = \int_{-\infty}^{+\infty} x^2 f_{(n)}(x)dx = \int_0^\theta \frac{n}{\theta^n}x^{n+1}dx = \frac{n}{n+2}\theta^2,$$

$$\text{则 } h(c) = E(T_c - \theta)^2 = \theta^2 \left(\frac{n}{n+2}c^2 - \frac{2n}{n+1}c + 1 \right), \text{所以当 } c = \frac{n+2}{n+1} \text{ 时, } h(c) \text{ 最小.}$$